

Grade 6 Accelerated
Unit 15
Lesson 5 Homework

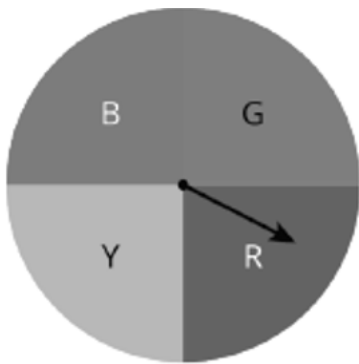
Name Answer Key
Date _____
Period _____

1. Explain the difference between theoretical and experimental probability of a simple experiment.

Theoretical is $\frac{\text{outcome}}{\text{total events in sample space}}$.

Experimental is $\frac{\text{times the event occurs}}{\text{total number of trials}}$.

Tyriek performed a simulation and spun the arrow on the spinner below. He recorded his results in the frequency table below.



Color	Frequency
Blue	7
Green	9
Red	4
Yellow	5

2. What is the theoretical probability of landing on red?

$$\frac{1}{4}$$

3. What is the experimental probability of landing on red?

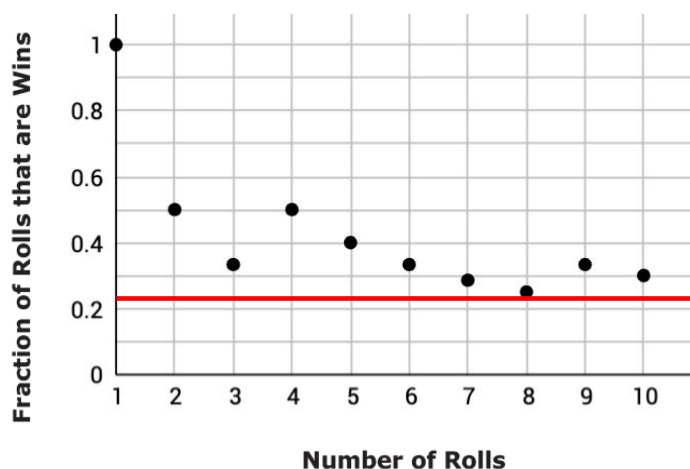
$$\frac{4}{7+9+4+5} = \frac{4}{25}$$

Arleni is playing a game where she wins if she rolls a 5 or a 6 on a standard number cube.

4. What is the theoretical probability that Alreni wins the game?

$$\frac{2}{6} \text{ or } \frac{1}{3}$$

She then performs the simulation 10 times. The graph below shows the fraction of games that are wins.



5. Estimate the probability of a roll winning this game based on the graph.

0.3

6. What type of probability is this?

experimental

7. Draw a horizontal line on the graph representing the theoretical probability of rolling a 5 or a 6.

8. After 10 rolls, what fraction of the total rolls were a win?

$\frac{3}{10}$

9. If Arleni rolled the number cube 10 more times, estimate the probability that she will roll a 5 or a 6.

$\frac{1}{3}$

10. If she were to roll the number cube 500, how many times would you expect for her to roll a 5 or a 6

$\frac{1}{3}(500) \approx 167$ times